

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	BE-CSE-AI	Date:	

Code of Conduct

*I **S.Muskan(192472182)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **S.Muskan**

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

- Analyze the semantic representations and identify the action-object relationships.
 - Determine which query contains semantic interpretation errors.
 - Evaluate how meaning representation affects chatbot decision accuracy.
 - Recommend improvements to enhance semantic understanding in telecom customer support systems.
2. First-Order Predicate Calculus for Smart Manufacturing.
An Industry 4.0 manufacturing plant monitors machines using semantic rules.
Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

- Represent the production data using First-Order Predicate Calculus.
 - Analyze which products are currently available.
 - Infer whether Gear production is affected by maintenance activities.
 - Evaluate the effectiveness of predicate logic in industrial decision-support systems.
3. Word Sense Disambiguation in E-Commerce Search Engines.
An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language

"Python course"	Snake	Programming Language
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Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

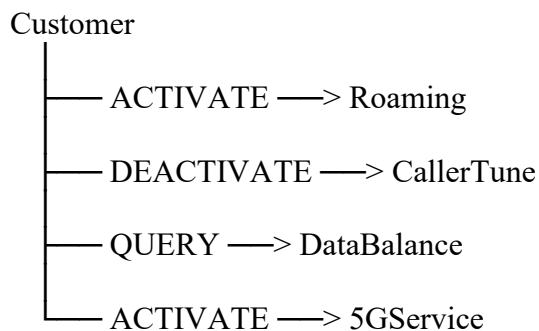
1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Q1. Semantic Representation in Customer Support Chatbots

1. Action–Object Relationships

Query	Semantic Representation	Action	Object	Interpretation
Q1	ACTIVATE(Roaming, Customer)	Activate	Roaming	Customer wants international roaming enabled
Q2	DEACTIVATE(CallerTune, Customer)	Deactivate	Caller Tune	Customer wants caller tune disabled
Q3	QUERY(DataBalance, Customer)	Query	Data Balance	Customer wants to know remaining data
Q4	ACTIVATE(5GService, Customer)	Activate	5G Service	Customer wants 5G enabled

Semantic structure:



2. Semantic Interpretation Error

Q2 contains the error.

- Actual intent: **Deactivate Caller Tune**
- Predicted intent: **Activate Caller Tune**

The semantic representation correctly identifies the intended action as:

DEACTIVATE(CallerTune, Customer)

However, the chatbot prediction changes the action from **DEACTIVATE** to **ACTIVATE**.

Q1, Q3, and Q4 have matching actual and predicted intents.

3. Effect on Chatbot Decision Accuracy

The chatbot correctly interprets **3 out of 4 queries**.

Accuracy = $\frac{3}{4} \times 100 = 75\%$

Therefore, the semantic interpretation error in Q2 reduces the decision accuracy to **75%**.

The error is significant because activating a service when the customer requested deactivation produces the **opposite action**.

Q1. SEMANTIC REPRESENTATION IN CUSTOMER SUPPORT CHATBOTS

1) Action–Object Relationships

Query ID	Customer Query	Semantic Representation	Action	Object
Q1	Activate international roaming for my number.	ACTIVATE(Roaming, Customer)	ACTIVATE	Roaming
Q2	Deactivate caller tune service.	DEACTIVATE(CallerTune, Customer)	DEACTIVATE	Caller Tune
Q3	Check my data balance.	QUERY(DataBalance, Customer)	QUERY	Data Balance
Q4	Enable 5G service.	ACTIVATE(5GService, Customer)	ACTIVATE	5G Service

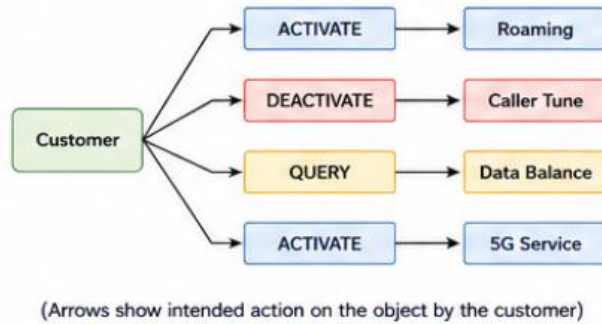
2) Semantic Interpretation Error

Q2 contains the error.
 Actual Intent : Deactivate Caller Tune
 Predicted Intent : Activate Caller Tune
 (Action misinterpreted)

3) Impact on Accuracy

Correct predictions = 3 out of 4
 Accuracy = $(3/4) \times 100 = 75\%$

4) Meaning Representation Network



4. Improvements

- Use explicit intent–action mapping such as ACTIVATE, DEACTIVATE, and QUERY.
- Detect negation and opposite-action words such as **deactivate**, **disable**, **cancel**, **stop**.
- Use context-aware intent classification.
- Validate the predicted action against the semantic representation before executing it.
- Add confirmation for potentially conflicting actions.

Conclusion: Accurate meaning representation is essential because a wrong action predicate can cause the chatbot to perform the opposite customer request.

Q2. First-Order Predicate Calculus for Smart Manufacturing

The production data on page 2 gives:

M1 $\rightarrow$ Active
M2 $\rightarrow$ Active
M3 $\rightarrow$ Maintenance
M4 $\rightarrow$ Active

1. First-Order Predicate Representation

Let:

- Active(x) = machine x is active
- Maintenance(x) = machine x is under maintenance
- Producing(x) = machine x is producing
- Produces(x,y) = machine x produces product y
- Available(y) = product y is available

Production facts:

Active(M1)
Active(M2)
Maintenance(M3)
Active(M4)

Using the given rules:

Active(x) $\rightarrow$ Producing(x)

Produces(x,y) $\wedge$ Active(x) $\rightarrow$ Available(y)

Maintenance(x) $\rightarrow \neg$ Producing(x)

2. Machines Currently Producing

From:

Active(x) $\rightarrow$ Producing(x)

we derive:

Active(M1) $\rightarrow$ Producing(M1)
Active(M2) $\rightarrow$ Producing(M2)
Active(M4) $\rightarrow$ Producing(M4)

From:

Maintenance(x) $\rightarrow \neg$ Producing(x)

we derive:

Maintenance(M3) $\rightarrow \neg$ Producing(M3)

Therefore:

M1 $\longrightarrow$ Producing
M2 $\longrightarrow$ Producing
M3 $\longrightarrow$ NOT Producing
M4 $\longrightarrow$ Producing

3. Which Products Are Available?

The rule is:

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

However, **no** $\text{Produces}(x,y)$ facts are provided in the dataset.

Therefore, it is not logically possible to identify a specific product as available from the given information.

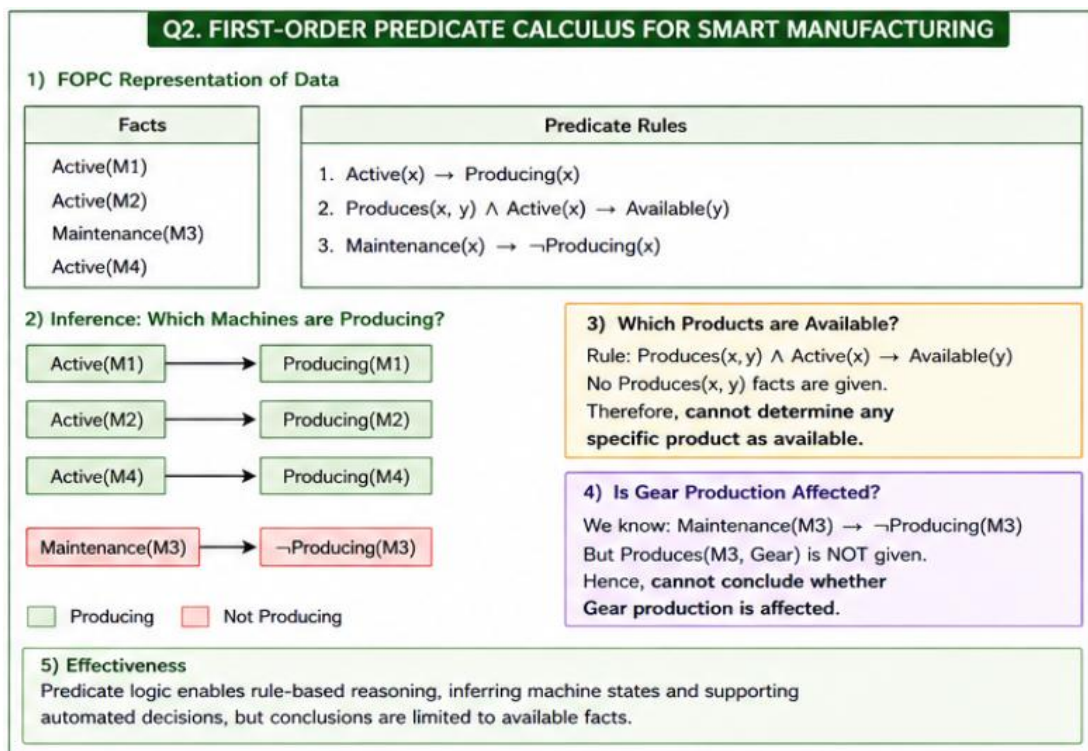
For example, if the missing fact were:

$\text{Produces}(\text{M1}, \text{Gear})$

then:

$\text{Produces}(\text{M1}, \text{Gear}) \wedge \text{Active}(\text{M1})$
 $\downarrow$
 $\text{Available}(\text{Gear})$

But without that fact, claiming that Gear is available would be unsupported.



4. Is Gear Production Affected by Maintenance?

The data establishes:

Maintenance(M3)

↓

¬Producing(M3)

Thus, **M3's production is affected by maintenance.**

However, the given data does **not** establish:

Produces(M3,Gear)

Therefore, we cannot definitively conclude that **Gear production** is affected.

Conditional conclusion:

If Produces(M3,Gear)

+

Maintenance(M3)

↓

Gear production is affected

5. Effectiveness of Predicate Logic

Predicate logic is effective because it:

- Represents machine states precisely.
- Allows rules to derive new production information.
- Clearly represents negative conditions such as ¬Producing(M3).
- Supports automated decision-making.

Conclusion: Predicate logic provides reliable rule-based reasoning, but conclusions are limited to the facts and rules supplied. Missing production relationships prevent specific product-level conclusions.

Q3. Word Sense Disambiguation in E-Commerce Search

The search context on page 3 provides clicked results that reveal the intended meaning of each ambiguous term.

1. Correct Word Sense

Query	Correct Sense	Evidence
Apple accessories	Technology Brand	Clicked result: iPhone Charger
Mouse wireless	Computer Device	Clicked result: Bluetooth Mouse
Java tutorial	Programming Language	Clicked result: Coding Lessons
Python course	Programming Language	Clicked result: Software Development Training

2. Semantic Cues-Apple accessories

"accessories"



electronic accessories



iPhone Charger



Technology Brand

Mouse wireless

"wireless"



wireless electronic device



Bluetooth Mouse



Computer Device

Java tutorial

"tutorial"



learning/coding context



Coding Lessons



Programming Language

Python course

"course"



educational/software context



Software Development Training



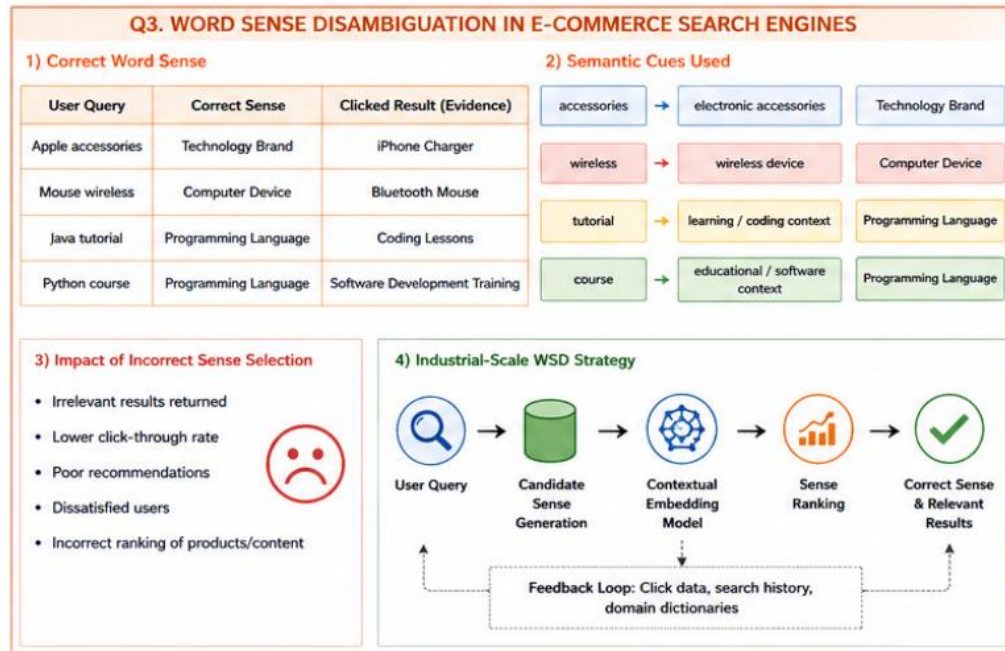
Programming Language

3. Impact of Incorrect Sense Selection

Incorrect WSD can cause:

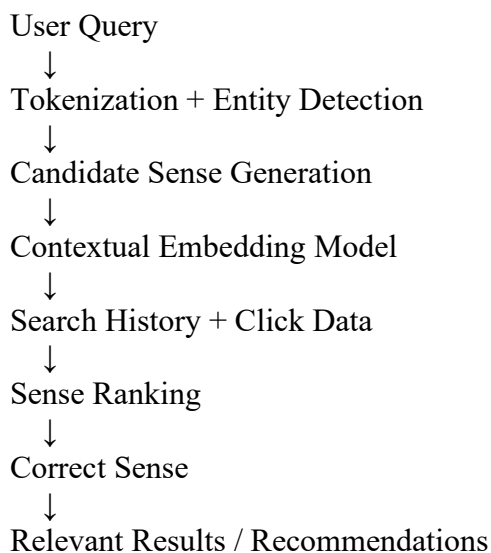
- Irrelevant search results.
- Lower click-through rate.
- Poor recommendations.
- Reduced user satisfaction.
- Incorrect product or content ranking.

For example, interpreting **"Python course"** as a snake would return irrelevant wildlife-related content instead of programming courses.



4. Industrial-Scale WSD Strategy

A practical WSD system can combine:



The system should use contextual embeddings, query history, click behaviour, domain-specific dictionaries, and continuously updated user/search feedback.

Conclusion: Context is the key signal for distinguishing ambiguous words. The provided click results strongly indicate the intended senses for all four queries.

Q4. Syntax-Driven Semantic Analysis in Healthcare

The document gives all four sentences as **Subject–Verb–Object** structures and provides semantic roles on page 3.

1. Relationship Between Syntax and Semantic Roles-Syntactic structure helps identify the participants involved in an action.

Example:

Doctor prescribed medicine to patient
| | | |
Subject Verb Object Recipient
| | | |
Agent Action Entity Recipient

Similarly:

Patient reported severe headache
| | |
Subject Verb Object
| | |
Experiencer Action Symptom

The parser identifies grammatical relationships first, which can then be mapped to semantic roles.

2. Evaluation of Assigned Roles

Entity	Given Role	Evaluation
Doctor	Agent	Appropriate
Medicine	Instrument	Not appropriate in the given sentence
Patient	Recipient	Appropriate for "prescribed medicine to patient"
Headache	Symptom	Appropriate

The main problem is **Medicine → Instrument**.

In:

"Medicine reduced blood pressure."

the medicine is the **cause/agent-like entity producing the reduction**, rather than an instrument used by another agent.

A better representation is:

Medicine
|
— CAUSES/AGENT —> Reduced Blood Pressure

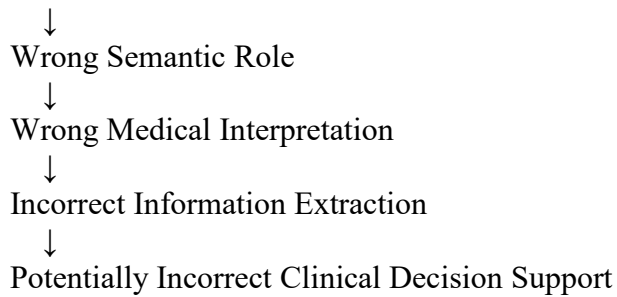
Also, for:-Patient reported severe headache.

the **Patient** is more naturally an **Experiencer**, while **Headache** is the **Symptom**.

3. Errors from Incorrect Parsing

Incorrect syntactic parsing can lead to:

Wrong Parse

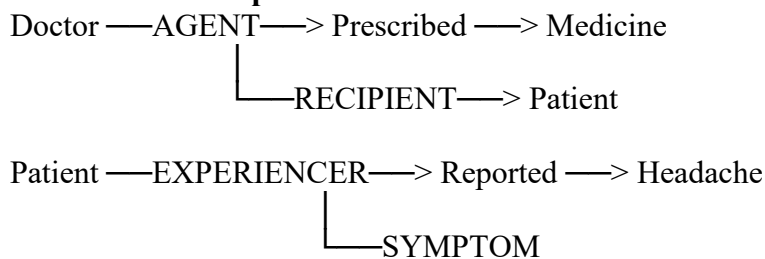


For example, incorrectly identifying the patient as the instrument or medicine as a recipient could distort the medical event representation.

4. Methods to Improve the System

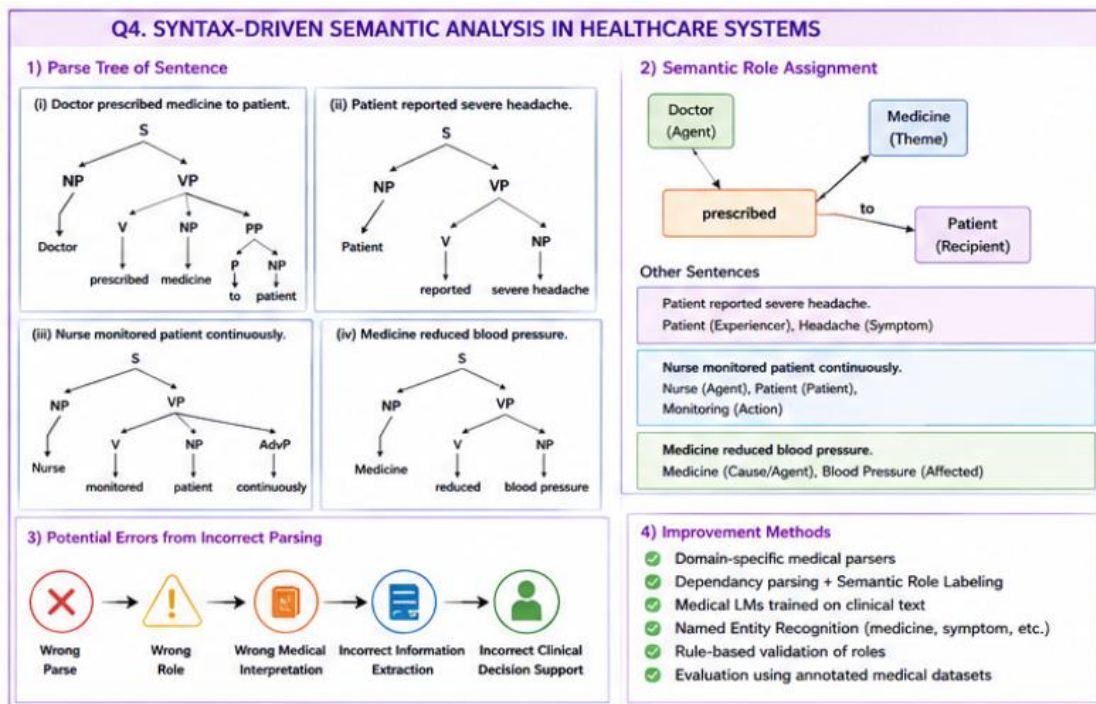
- Use domain-specific medical parsers.
- Combine dependency parsing with semantic role labeling.
- Use medical language models trained on clinical text.
- Apply entity recognition for medicines, symptoms, diseases, and patients.
- Use rule-based validation for impossible or unlikely role assignments.
- Evaluate the system using annotated medical datasets.

Semantic Role Representation



Medicine —CAUSE—> Reduced —> Blood Pressure

Conclusion: Syntax provides the structural foundation for semantic interpretation, but syntactic roles must be validated against the actual meaning of medical events. The **Medicine = Instrument** assignment is the key semantic error in the supplied data.



Overall Conclusion

The four scenarios demonstrate that NLP systems depend on **accurate meaning representation, logical inference, contextual word-sense selection, and syntax-to-semantics mapping**. The data shows:

- **Q1:** One chatbot intent error gives **75% accuracy**.
- **Q2:** Machine states can be inferred, but product availability cannot be determined without Produces(x,y) facts.
- **Q3:** Context correctly disambiguates all four e-commerce queries.
- **Q4:** Most semantic roles are reasonable, but **Medicine as Instrument** is inappropriate for the given sentence.

Rubrics For Calculation:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

